stats_hw6

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2025-03-21

Problem 23

Complete and make a boxplot showing the oxygen distribution for each bone.

```r
library(dplyr)
```

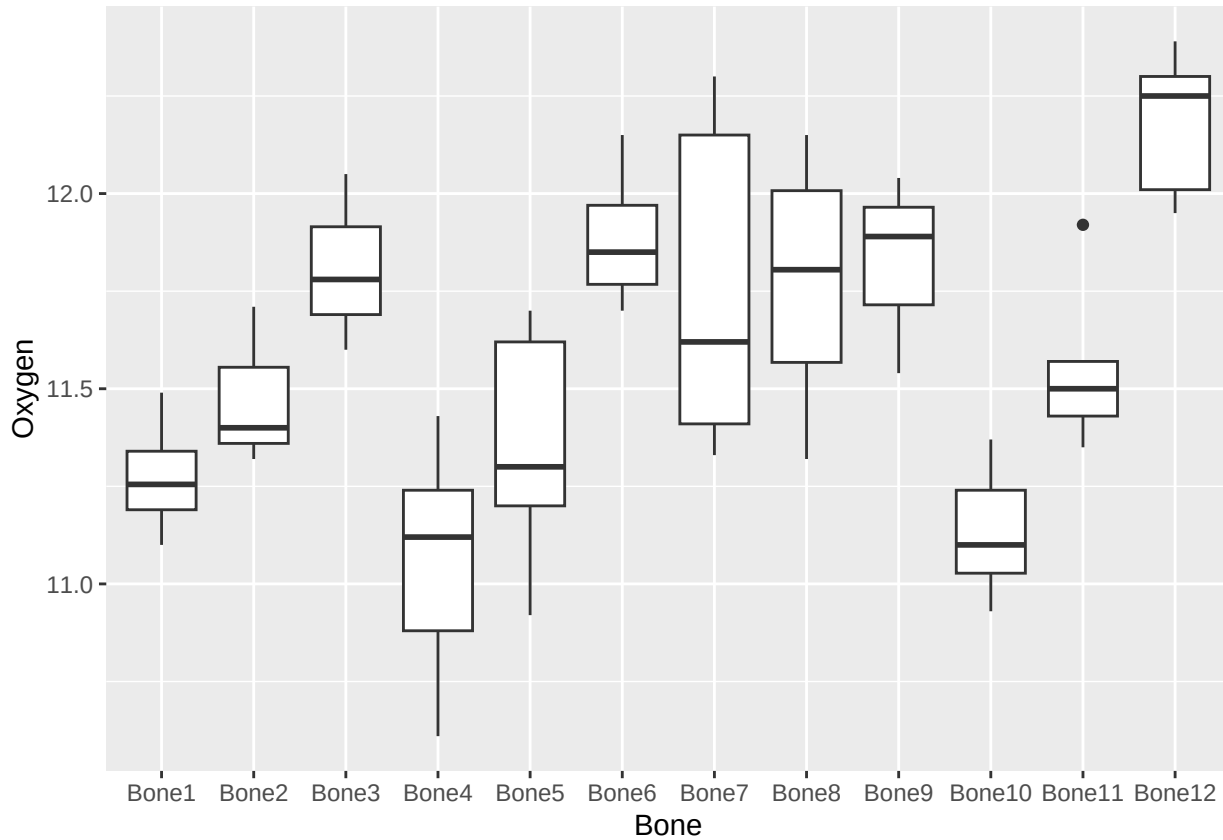
```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(Sleuth3)
library(forcats)
library(stringr)

ex0523 |>
  mutate(Bone = fct_reorder(Bone, as.numeric(str_extract(Bone, "\\d+")))) |>
  ggplot(aes(x=Bone, y=Oxygen)) + geom_boxplot()
```




```r
lm(Oxygen ~ Bone, data=ex0523) |> anova()
```

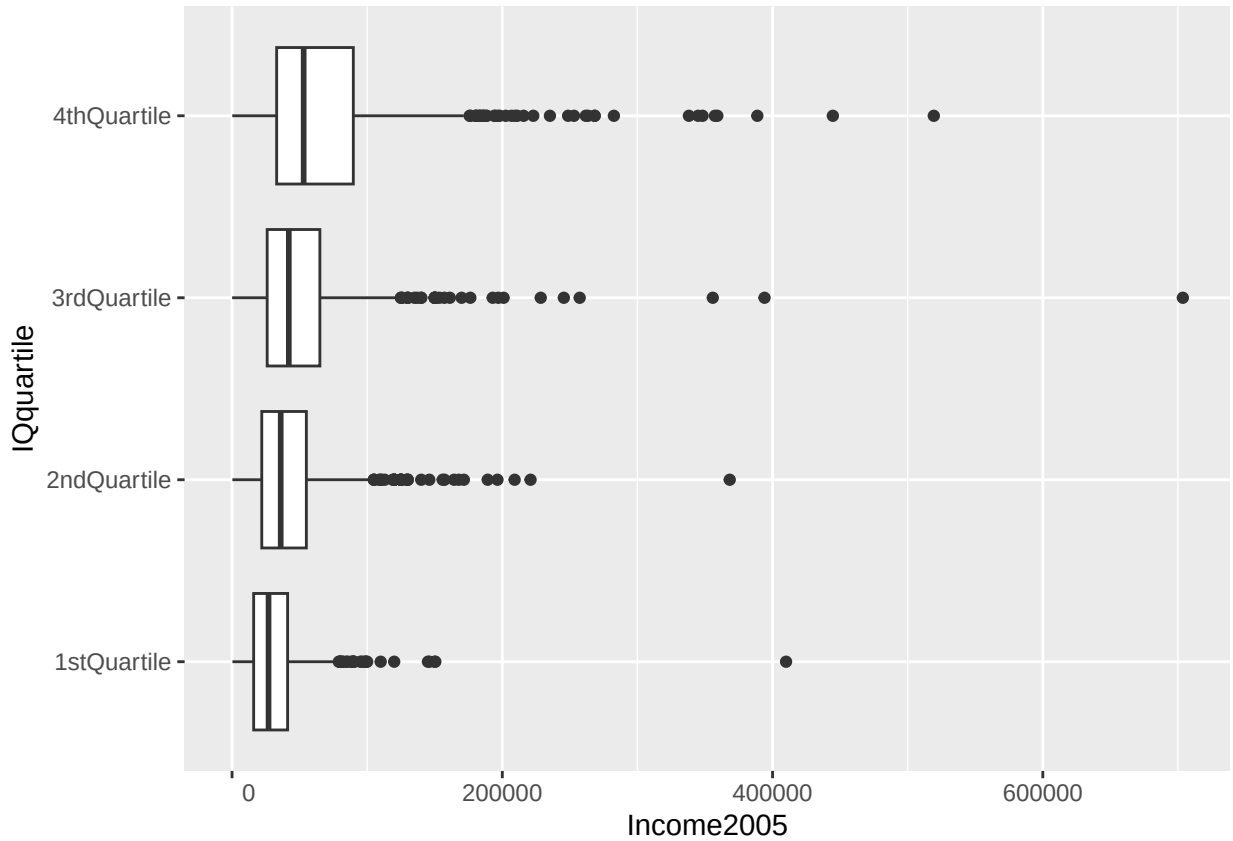
```
## Analysis of Variance Table
##
## Response: Oxygen
##           Df Sum Sq Mean Sq F value   Pr(>F)
## Bone      11 6.0675 0.55159  7.4268 9.73e-07 ***
## Residuals 40 2.9708 0.07427
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Because there is strong statistical significance, we can somewhat preemptively conclude that the Tyrannosaurus Rex was warm-blooded.

Problem 24

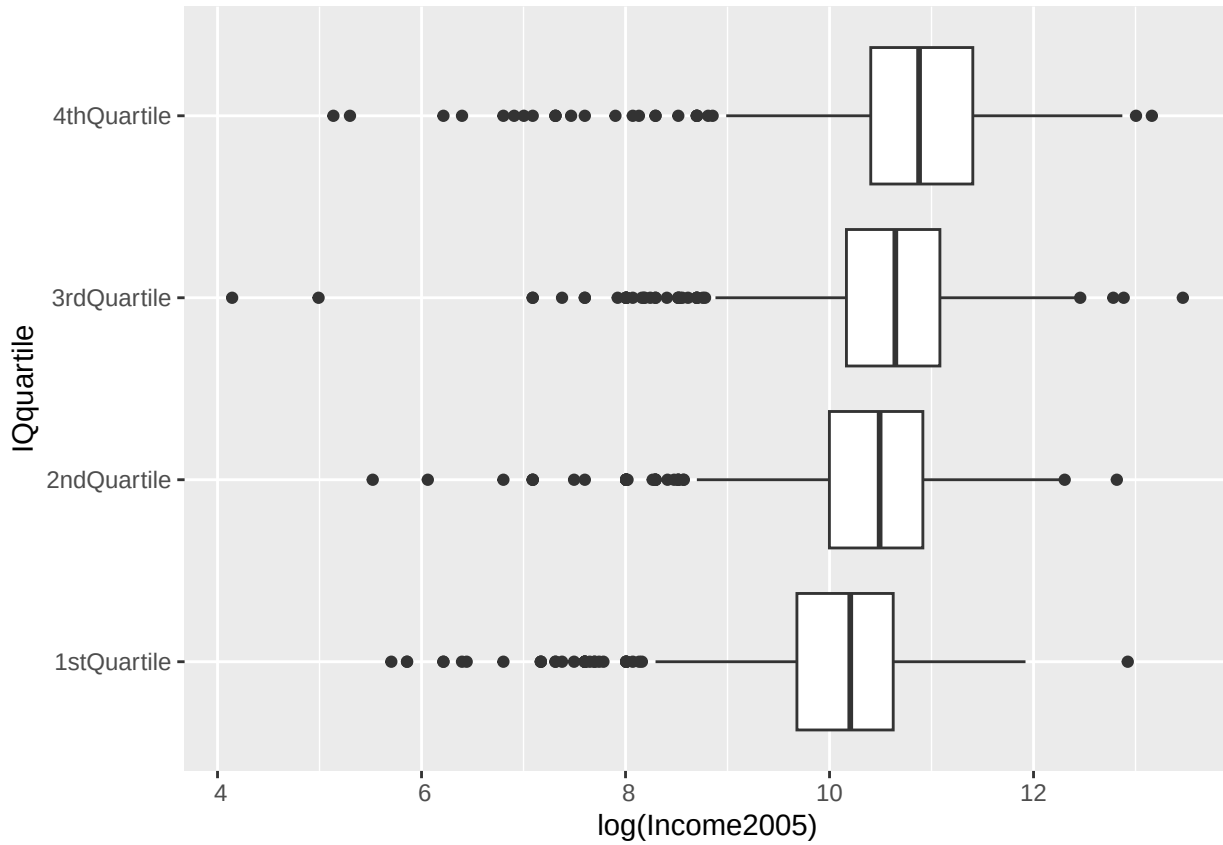
Make a boxplot showing distribution of Income 2005 for each IQ quartile. Observe inconstant variance - higher incomes have larger spreads. Make a new boxplot showing the log of Income2005, and use the logarithm of Income2005 in your ANOVA.

```r
ex0524 |> ggplot(aes(x=Income2005, y=IQquartile)) + geom_boxplot() +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE))
```




```
# Larger variances in larger income groups

ex0524 |> ggplot(aes(x=log(Income2005), y=IQquartile)) + geom_boxplot() +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE))
```




```r
income <- ex0524 |>
  mutate(IQquartile = recode(IQquartile,
                             "1stQuartile" = "1",
                             "2ndQuartile" = "2",
                             "3rdQuartile" = "3",
                             "4thQuartile" = "4")) |>
  mutate(IQquartile = as.numeric(IQquartile))  #converting quartiles to numeric

lm(IQquartile ~ log(Income2005), data=income) |> anova()
```

```
## Analysis of Variance Table
##
## Response: IQquartile
##                   Df  Sum Sq Mean Sq F value    Pr(>F)
## log(Income2005)    1  240.13 240.127  207.37 < 2.2e-16 ***
## Residuals       2582 2989.87   1.158
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
#differences in 4thQuartile to 1stQuartile
first <- ex0524 |> filter(IQquartile == "1stQuartile") |>
  summarize(mean(Income2005))
fourth <- ex0524 |> filter(IQquartile == "4thQuartile") |>
  summarize(mean(Income2005))
diff <- fourth - first
print(diff)
```

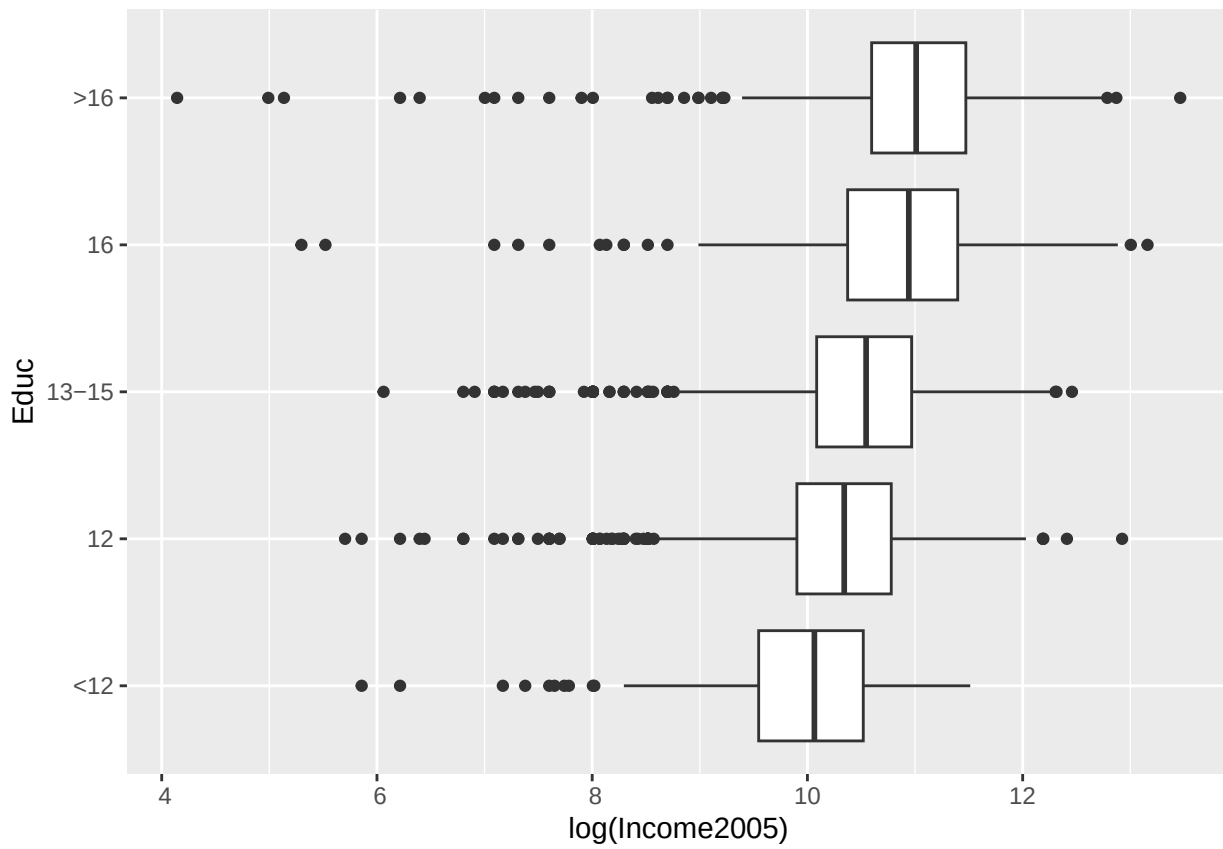
```
##   mean(Income2005)
## 1         37892.55
```

The mean income differences between the first and fourth quartile of IQ is $37,892.55.

Problem 25

Make a boxplot showing the log of Income2005 for each group, and reorder the education groups so that "<12" is the first group in the plot, as it should be. Finish the problem using the log(Income2005).

```
ex0525 |> mutate(Educ = forcats::fct_relevel(Educ, "<12")) |>
  ggplot(aes(x=log(Income2005), y=Educ)) + geom_boxplot() +
  scale_x_continuous(labels = function(x) format(x, scientific = FALSE))
```




```
education <- ex0525 |>
  mutate(Educ = recode(Educ,
                              "<12" = "1",
                              "12" = "2",
                              "13-15" = "3",
                              "16" = "4",
                              ">16" = "5")) |>
  mutate(Educ = as.numeric(Educ))  #converting educ to numeric
```

```r
lm(Educ ~ log(Income2005), data=education) |> anova()
```

```
## Analysis of Variance Table
##
## Response: Educ
##                   Df Sum Sq Mean Sq F value    Pr(>F)
## log(Income2005)    1  217.8 217.841   115.2 < 2.2e-16 ***
## Residuals       2582 4882.7   1.891
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
everyoneelse <- ex0525 |> filter(Educ != "<12") |> summarize(mean(Income2005))
lesstwelve <- ex0525 |> filter(Educ == "<12") |> summarize(mean(Income2005))

diff2 <- everyoneelse - lesstwelve
print(diff2)
```

```
##   mean(Income2005)
## 1         22288.64
```

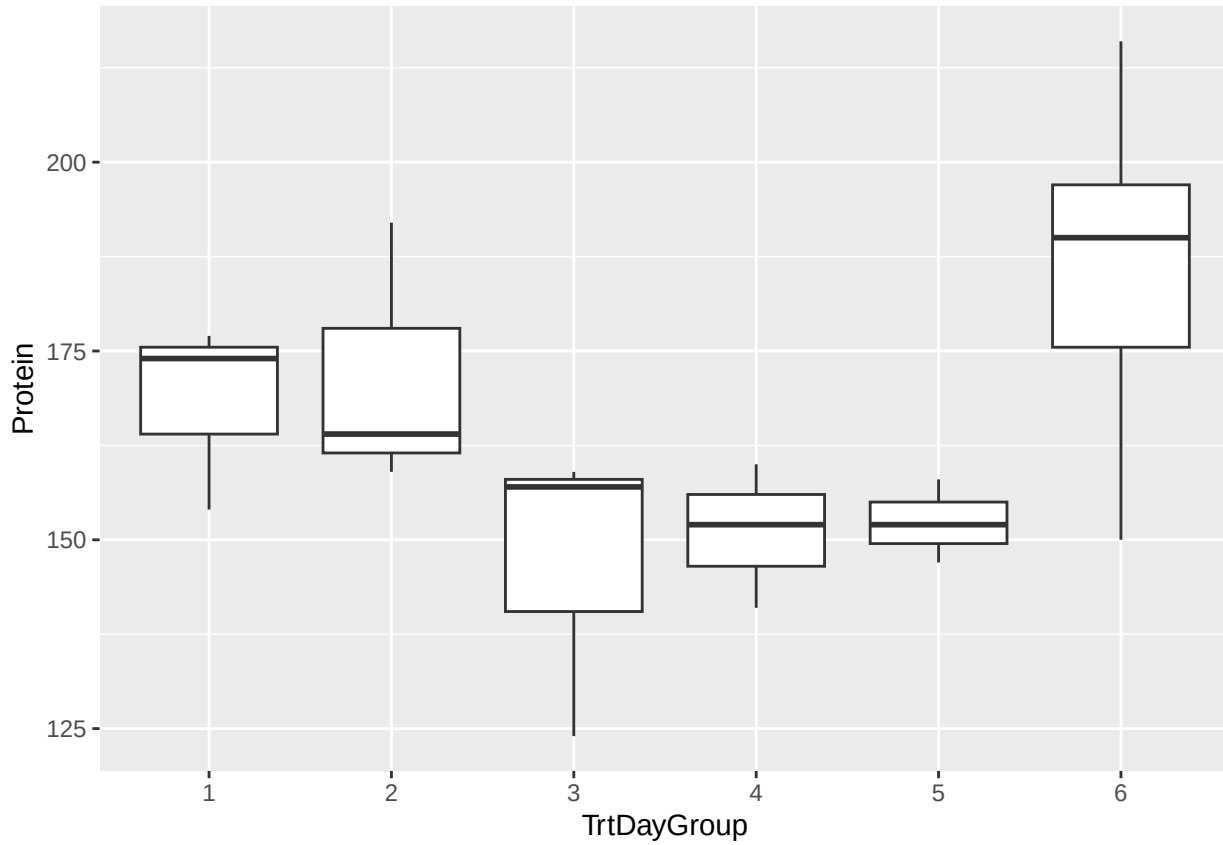
Problem 18

part a: Estimate means for the model with six independent samples, one for each treatment. Determine the residuals and plot them versus the estimated means. Plot the residuals versus the day on which the investigations was conducted. Is there any indication that the methods of this chapter are not appropriate? (Dr. Clair: just conduct anova f-test to see whether the six treatments have different means)

Make a boxplot showing all ten TrtDayGroup protein distributions, and color by Treatment so it's clear which boxes go with which treatments. Re-order the groups so that they appear in numerical order.

```r
mouse6 <- ex0518 |> mutate(TrtDayGroup = recode(TrtDayGroup,
                            "Group1" = "1",
                            "Group2" = "2",
                            "Group3" = "3",
                            "Group4" = "4",
                            "Group5" = "5",
                            "Group6" = "6",
                            "Group7" = "6",
                            "Group8" = "6",
                            "Group9" = "6",
                            "Group10" = "6",)) |>
  mutate(TrtDayGroup = factor(TrtDayGroup, levels = c("1", "2", "3", "4", "5", "6")))

ggplot(mouse6, aes(x = TrtDayGroup, y = Protein)) + geom_boxplot()
```



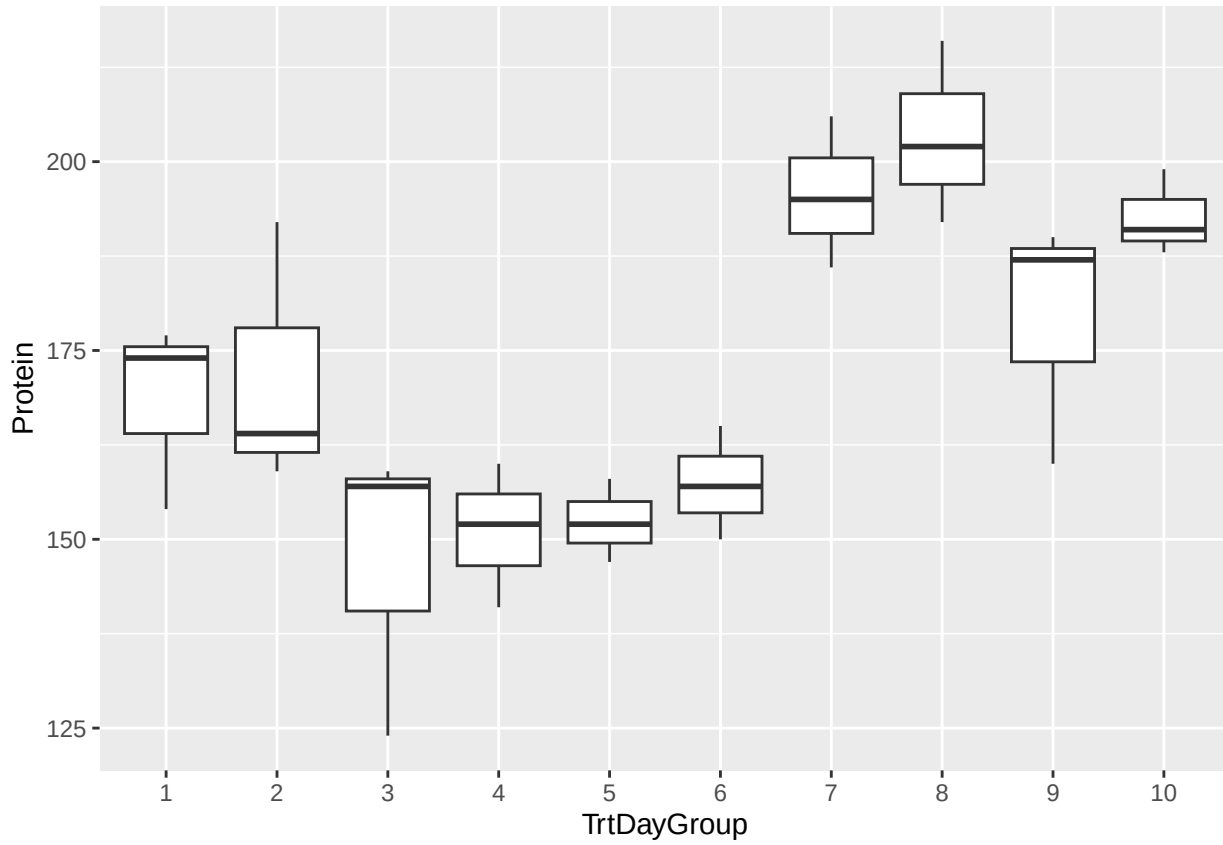

```
mean6mouses <- lm(Protein ~ TrtDayGroup, data=mouse6)
anova(mean6mouses)
```

```
## Analysis of Variance Table
##
## Response: Protein
##             Df Sum Sq Mean Sq F value   Pr(>F)
## TrtDayGroup  5 7222.5 1444.51  4.8827 0.003196 **
## Residuals   24 7100.3  295.84
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

part b:

```
ex0518 |> mutate(TrtDayGroup = recode(TrtDayGroup,
                              "Group1" = "1",
                              "Group2" = "2",
                              "Group3" = "3",
                              "Group4" = "4",
                              "Group5" = "5",
                              "Group6" = "6",
                              "Group7" = "7",
                              "Group8" = "8",
                              "Group9" = "9",
                              "Group10" = "10",)) |>
```

```r
  mutate(TrtDayGroup = forcats::fct_relevel(TrtDayGroup, "10", after=9)) |>
      ggplot(aes(x=TrtDayGroup, y=Protein)) + geom_boxplot()
```




```r
mean10mouses <- lm(Protein ~ TrtDayGroup, data=ex0518)
anova(mean10mouses)
```

```
## Analysis of Variance Table
##
## Response: Protein
##             Df  Sum Sq Mean Sq F value    Pr(>F)
## TrtDayGroup  9 11147.5 1238.61  7.8014 7.154e-05 ***
## Residuals   20  3175.3  158.77
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

part c:

```r
anova(mean6mouses, mean10mouses)
```

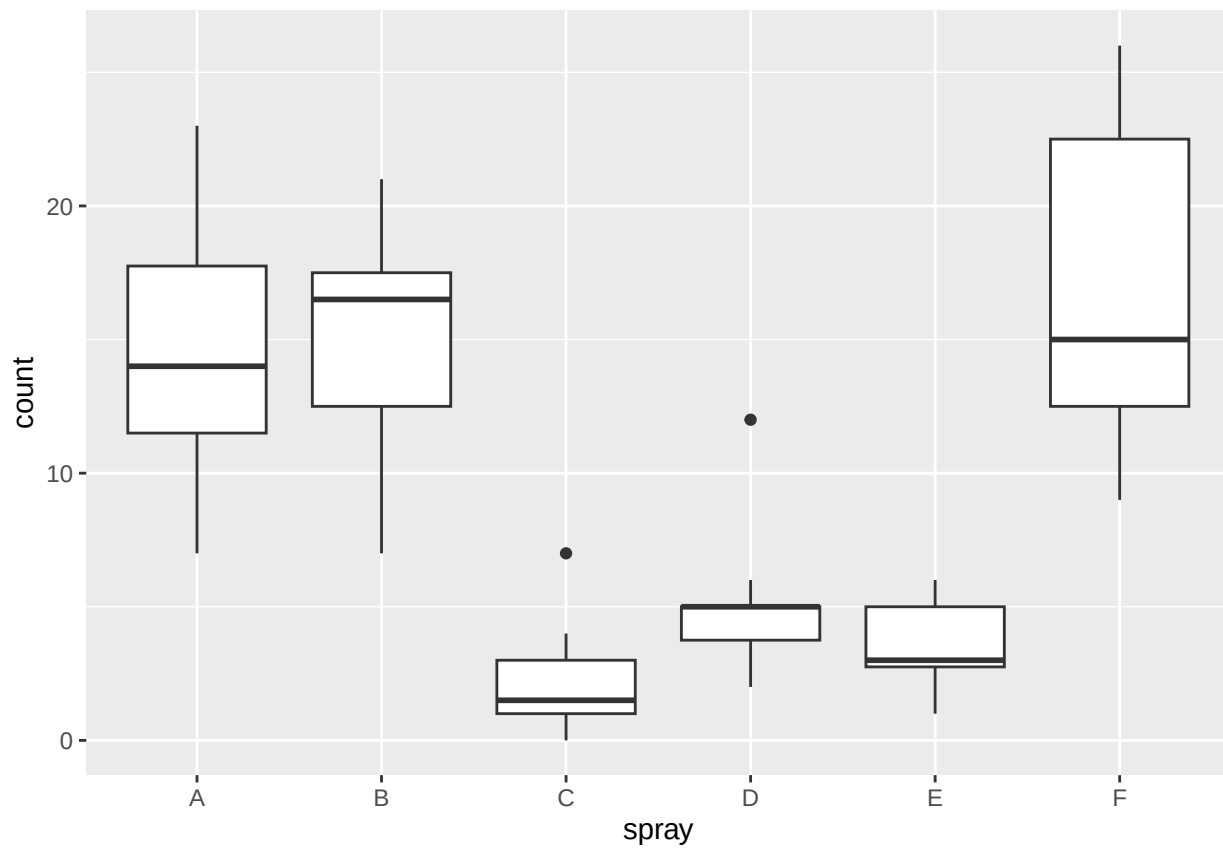
```
## Analysis of Variance Table
##
## Model 1: Protein ~ TrtDayGroup
## Model 2: Protein ~ TrtDayGroup
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
```

```
## 1     24 7100.3
## 2     20 3175.3  4    3924.9 6.1803 0.002089 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Problem InsectSprays

part a: plot the data.

```
InsectSprays |> ggplot(aes(x=spray, y=count)) +geom_boxplot()
```



Sprays C, D, and E seem to be more effective.

part b: use a one-way anova to test if the three effective sprays have the same mean.

```
threebugs <- InsectSprays |> filter(spray %in% c("C", "D", "E"))
lm(count ~ spray, data=threebugs) |> anova()
```

```
## Analysis of Variance Table
##
## Response: count
##           Df  Sum Sq Mean Sq F value   Pr(>F)
## spray      2  48.167 24.0833  5.4873 0.008763 **
## Residuals 33 144.833  4.3889
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A mean of 4.3889 makes sense looking at the boxplot, and with a strong significance.

part c: use a one way anova to test if the three less effective sprays have the same means.

```
otherthreebugs <- InsectSprays |> filter(spray %in% c("A", "B", "F"))
lm(count ~ spray, data=otherthreebugs) |> anova()
```

```
## Analysis of Variance Table
##
## Response: count
##           Df Sum Sq Mean Sq F value Pr(>F)
## spray      2  28.67  14.333  0.5435 0.5858
## Residuals 33 870.33  26.374
```

Because this result is not statistically significant, the mean does not make sense.

part d: Would it be appropriate to use a one-way anova on the entire data set? Why or why not?

I do not believe a one way anova would be appropriate on this whole data set as as a one way anova means there is a independent variable to compare all the other means to.

part e: use oneway.test to test the null hypothesis that all of the mean insect counts are the same for the various sprays versus the alternative that they are not, at the signifiance level=0.01 level.

```
oneway.test(count ~ spray, data=InsectSprays)
```

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  count and spray
## F = 36.065, num df = 5.000, denom df = 30.043, p-value = 7.999e-12
```

A p-value is extremely significant, which means I can safely conclude the differences in the means are not due to chance and reject the null hypothesis.